2 The Theory and Practice of Market Capitalism

All the bad things you hear about markets are true: unemployment, inflation, inequalities of income and wealth, monopoly power, negative externalities, and insufficiently supplied public goods. You know, there is only one thing that is worse than the market, and that is no market.
—Csaba Csáki, Rector of Economics, Budapest University of Economic Sciences (formerly Karl Marx University), speech, August 1990

INTRODUCTION

The vast changes sweeping through the world economy have focused attention upon the nature of the market capitalist economic system, the system that is the goal of many reformers in power in the former communist countries. Even many predominantly market capitalist economies are making efforts to move toward a purer version of this system, although this effort may be slowing down in some places. It seems, as Fukuyama argues, to be the victorious universal ideology of the world.

We have never seen a pure version of the market capitalist system anywhere in history, nor are we likely to. The U.S. and British economies in the middle to late nineteenth century were probably the closest achievement of pure market capitalism. These economies represented the culmination of a historical line of development that, originating in the murky mists of time, formed a coherent system in the 1200s in northern Italy and Flanders with the invention of modern accounting and accelerating urbanization, and transformed itself into a dominating structure with the **Industrial Revolution** in Great Britain in the late eighteenth century. These processes also involved an evolution of institutions that allowed capitalist markets to operate without excessive transactions costs.[1] But even at the apogee of pure market capitalism in the nineteenth century, governments intervened in many ways, from trade protectionism to subsidizing the building of transportation infrastructure to maintaining military forces.

These economies exhibited both the advantages and difficulties of unfettered market capitalism. They experienced enormous technological advances and growth as they underwent the Industrial Revolution. Even those critics of market capitalism, Karl Marx and Friedrich Engels, recognized the enormous ability of the system to "revolutionize the means of production" in ways unprecedented in world history.

However, both the American and British economies experienced large macroeconomic fluctuations with serious downturns in the 1870s and 1890s and increasingly unequal distributions of income associated with increasing concentrations of industrial monopoly

1. The New Institutional School of economics especially emphasizes the role of the historical evolution of legal frameworks that support the enforcement of contracts and property rights and the reduction of transactions costs more generally as fundamental to the rise of market capitalism. Douglass North has particularly studied this element in the historical rise of market capitalism in Western Europe (Douglass C. North and Robert Paul Thomas, *The Rise of the Western World: A New Economic History* [Cambridge: Cambridge University Press, 1973]).

power. After 1900 these problems triggered substantial movements toward greater government involvement in both economies, in the United States especially after the Great Depression of the 1930s.

Today three economies that come closest to the ideal of pure laissez-faire market capitalism are Hong Kong, Singapore, and New Zealand.[2] The first two have highly successful records in many ways. In recent decades both Hong Kong and Singapore have enjoyed among the highest growth rates in the world along with very low unemployment. We shall consider the case of Hong Kong in chapter 15, but for now note that, much like Japan, there has been indicative planning by the Hong Kong authorities. There has also been state ownership of land with long-term leases. Hong Kong is not an independent country, having been a crown colony of Great Britain before 1997 and a part of the People's Republic of China since then. Despite interventions in the stock market during the Asian financial crisis of 1997 and despite coming under the control of the Communist government in Beijing, Hong Kong has largely retained its mostly laissez-faire system, marked by very low taxes, free trade, no state enterprises, and few business regulations. In Singapore, there are more state-owned enterprises that supply about 28 percent of government revenues. Neither Hong Kong nor Singapore is a properly functioning democracy, with Singapore having a reputation for political authoritarianism despite its free-market economy.

Whereas Hong Kong has long had a very laissez-faire economy, New Zealand has become such an economy only since 1984 with a major wave of tariff reductions, privatization of formerly state-owned enterprises, more general deregulation, and reduced social safety nets. Although the growth rate increased after these reforms occurred, New Zealand's performance has not been as impressive as that of Hong Kong or Singapore. New Zealand had an annual GDP growth rate of 1.8 percent during the 1990s and as of 2000 was only 23rd in real per capita income in the world. It also has a striking degree of income inequality, as can be seen in table 1-1, where its quintile ratio of 17.4 is the highest among all high HDI nations. Thus, it is unsurprising that a reaction against the cuts in social safety nets occurred after 1999. New Zealand has undone some of its moves toward laissez-faire capitalism by renationalizing the accident compensation scheme, raising the top marginal income tax rate, and increasing environmental restrictions on natural resource use. This shows the limits of laissez-faire in political democracies. Thus, despite their high EFI ratings, neither Hong Kong nor Singapore nor New Zealand is a pure example of a laissez-faire economy.

2. Ratings of economic freedom in 155 countries can be found in Gerald P. O'Driscoll, Jr., Kim R. Holmes, and Melanie Kirkpatrick, *2001 Index of Economic Freedom* (Washington, D.C., and New York: Heritage Foundation and the *Wall Street Journal,* 2002). Hong Kong rates as the most economically free country in the world, with Singapore second and New Zealand third, according to this volume. The United States is tied for fourth place with Estonia, Ireland, Luxembourg, and the Netherlands. North Korea is 155th.

Some very poor less developed countries have smaller state sectors relative to GDP than these three—for example, desperately poor Sierra Leone.[3] But most of these nations also have poorly developed markets and little modern industrial capitalism. Among the larger countries, the economy most oriented to market capitalism is probably the United States, despite its substantial increase in government intervention since the 1930s, discussed in more detail in chapter 5.

It is the **dynamic efficiency** or technological dynamism of market capitalism that is its greatest appeal to countries seeking to emulate its success. But this dynamism has come through the macroeconomically destabilizing process of **creative destruction,** as described by Joseph Schumpeter.[4] With respect to static efficiency, market capitalism can be seen as possessing significant advantages, although Adam Smith strongly argued for both market capitalism's dynamic and static advantages. Both of these efficiencies require market capitalism to operate within an institutional framework that allows transactions and contracts to take place and be enforced without excessive cost.

THE THEORETICAL EFFICIENCY OF MARKET CAPITALISM

Why do some of the world's highest real per capita income countries have economies that are market capitalist, such as the United States? Probably the strongest reason is the general ability of markets to allocate goods and resources *efficiently* through the law of supply and demand. This general ability is summarized in the following theorem: *A complete, competitive, full-information general equilibrium is efficient*.

To understand this theorem, its implications, and its limitations, it is necessary to know what the terms in it mean. **Complete** means that for any good or service that affects someone's utility, there is a market. **Competition** means that there are many buyers and sellers with free entry and exit, there are well-defined homogeneous goods and services, and no individual supplier has any control over the price in his or her market. **Full information** means that all agents in the economy know everything about consumer preferences, production technologies, prices, and anything else they might need to know in deciding how to act. **General equilibrium** means that every single market is in equilibrium in the sense that the quantity supplied equals the quantity demanded of the good or service in question. If only a few markets are in equilibrium, this is **partial equilibrium. Efficiency** means Pareto optimality, after the Italian economist Vilfredo Pareto. No one in the economy can be made

3. There is an argument known as **Wagner's Law** that asserts that as an economy's income rises the relative size of its state sector expands.

4. Joseph A. Schumpeter, *The Theory of Economic Development* (Cambridge, Mass.: Harvard University Press, 1934), originally published in German in 1911.

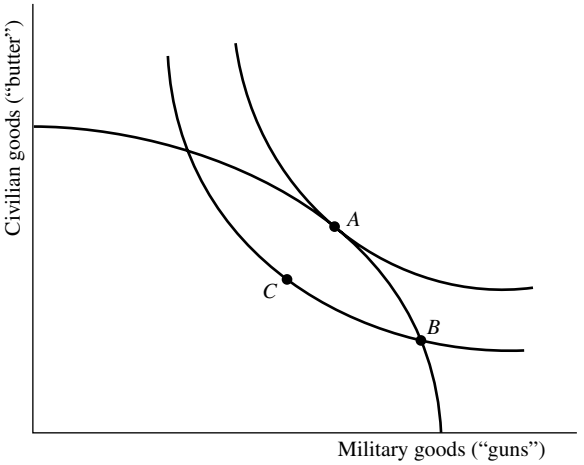


Figure 2-1
Production possibilities frontier.

better off without making someone else worse off. If someone can be made better off without making someone else worse off, then the economy is not producing as much as possible of what people want. But if Pareto optimality holds, no more of what people want can be produced; all that can be done is to reshuffle existing goods and services between people.

Thus the economy is on its **production possibilities frontier** (PPF), defined as the set of maximum possible output combinations the economy can produce given its resources and technology. But not all points on the PPF are efficient; some may represent combinations of goods and services that people do not want. The former Soviet economy may have been on its PPF but was thought to produce too much military equipment and not enough civilian consumer goods.

An efficient economy must be a fully employed economy. Otherwise, it would be lying inside the PPF because the unemployed could presumably produce at least some more of one good without reducing the output of another. Thus microeconomic efficiency implies macroeconomic full employment.

Figure 2-1 shows a PPF with point *A* being Pareto optimal at the tangency between the social indifference curve[5] and the PPF, while *B* lies at a lower level of utility on the PPF and *C* represents a point of unemployed factors located inside the PPF.

5. To draw such a curve implies that everyone has identical preferences, a strong simplifying assumption. For a discussion of the difficulties in forming such a social welfare function democratically when people's preferences differ, see Kenneth J. Arrow, *Social Choice and Individual Values,* 2nd ed. (New York: Wiley, 1963).

This concept of efficiency says nothing about income distribution. An economy might be efficient according to Pareto if it has completely equal distribution of income or if one person has everything and everyone else is starving to death, just as long as no one can be made better off without making someone else worse off, the necessary condition for Pareto optimality. The tendency to unequal wealth and incomes under market capitalism has been one of the major arguments against it raised by socialist critics. But this is a criticism distinct from the issue of economic efficiency. The existence of a general equilibrium presupposes a prior distribution of wealth, with the income distribution arising from the general equilibrium itself.

Returning to the main argument, why is a complete, competitive, full-information, general equilibrium efficient? The underlying intuition dates to Adam Smith's invocation of the **invisible hand** of the market working across all sectors to allocate goods in a way that maximizes the "wealth of nations," although Smith had no formal concept of a general equilibrium, which was first defined by Léon Walras in 1874. Although Pareto argued for the link between general equilibrium and efficiency in 1909, it was Kenneth Arrow and Gérard Debreu in 1954[6] who presented a formal mathematical proof of both the existence and efficiency of competitive general equilibrium.

The efficiency of competitive general equilibrium is most easily seen by looking at a partial equilibrium case, the outcome in a single market. Figure 2-2 shows a typical competitive market with an upward-sloping supply curve (S) and a downward-sloping demand curve (D). The solutions for three different prices are shown: P_1 above equilibrium, P^* at equilibrium, and P_2 below equilibrium.

At P_1 suppliers produce more than they would at equilibrium but buyers buy less than they would at equilibrium, resulting in a surplus equal to the quantity produced that no one wants to buy. The amount that is both produced and sold is less than occurs at equilibrium. At P_2 demanders buy more than they would at equilibrium but suppliers produce less than they would at equilibrium, resulting in a shortage equal to the quantity buyers want that has not been produced. Again, the amount that is both produced and sold is less than occurs at equilibrium.

It is at the equilibrium price that the maximum amount will be both produced and sold and thus actually consumed by the public. This argument extends to all markets in the general equilibrium case. Thus, to maximize the amount of all goods available for consumption, every market should be in equilibrium.

6. For a discussion of this history and these arguments, see Kenneth Arrow and Frank Hahn, *General Competitive Analysis* (San Francisco: Holden-Day, 1971), chap. 1.

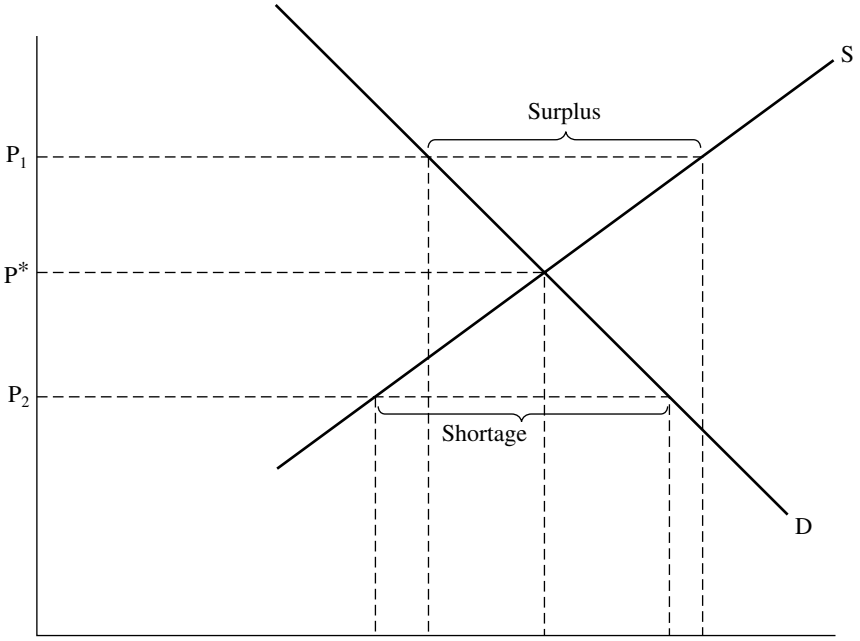


Figure 2-2
Equilibrium of competitive supply and demand.

LIMITS TO THE EFFICIENCY OF LAISSEZ-FAIRE MARKET CAPITALISM

Monopoly Power

No one should underestimate the power and significance of the preceding efficiency theorem as it is used to support market capitalism. However, no one should be fooled into thinking that absolute laissez-faire market capitalism fulfills the conditions of the theorem and is, therefore, automatically efficient. In general, laissez-faire market capitalism will not be completely efficient; this situation is called the problem of **market failure.**

One condition of efficient equilibrium is that it is competitive. Monopoly power is a source of inefficiency and can arise in a laissez-faire economy. This type of market failure happened in the United States at the end of the 1890s, culminating in a great concentration of monopoly power that was attacked by President Theodore Roosevelt, the "Trust Buster," after 1901.

Figure 2-3 compares a pure monopoly outcome with a perfectly competitive solution. The monopolist will maximize profits by setting marginal cost (MC) equal to marginal

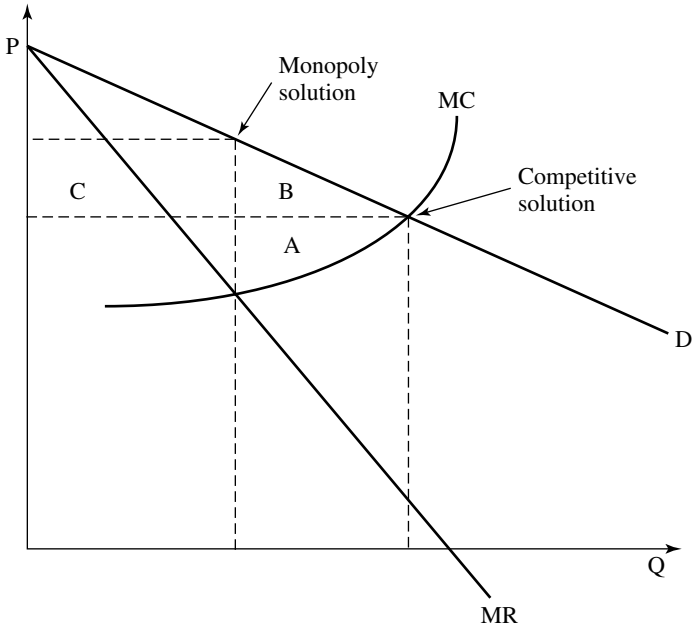


Figure 2-3
Monopoly and perfect competition compared.

revenue (MR), whereas if the industry is perfectly competitive, the equilibrium will be at P = MC, where MC intersects the demand curve. The monopolist will produce less and charge a higher price than would the competitive industry. Triangle *A* shows lost **producer's surplus** (net income for the producer), and triangle *B* shows lost **consumer's surplus** (net utility) caused by the reduced production. Despite losing triangle *A*, the producer gains on net because she obtains the larger rectangle *C* of **monopoly rent** from the consumer as a result of the higher price.[7] This market failure would not occur in a perfectly competitive situation.

Despite this general critique caused by monopoly power, some qualifying caveats are in order. The first is the situation of **natural monopoly,** characterizing an industry with economies of scale (declining long-run average costs) even at a level of output equal to total

7. Probably the basis for political support for antitrust policy in the United States is this *rectangle of redistribution* (Tullock rent) rather than the missing *triangles of inefficiency* (Harberger triangles). Consumers get angry when "ripped off" by a visibly rich monopolist. The missing triangles are invisible because they do not exist, and according to some estimates are only on the order of 1 percent of GDP in the United States anyway.

market demand. If a single firm can produce the total market demand at a lower cost than can many firms, then it is desirable to enjoy these efficiencies of large-scale production.

The existence of natural monopoly presents an inevitable trilemma. Either laissez-faire can be followed, in which case consumers will experience losses, or the government can regulate the monopoly, as state governments traditionally did in the United States with respect to electric utilities prior to the recent interstate deregulation movement. But regulation often leads to inefficiencies, such as overinvestment in capital stock, when such firms are guaranteed a particular rate of return on invested capital. Or the natural monopoly can be run by the government, which entails socialism with all its bureaucratic inefficiency. This is the system in much of Western Europe for electric utilities.

Another caveat involves technological dynamism. It is argued that more competitive industries will be more technologically progressive. But if research and development (R and D) exhibit economies of scale, then a large monopolist with large monopoly profits may generate more R and D if it can be sufficiently motivated, as argued by Joseph Schumpeter in *Capitalism, Socialism and Democracy* (1943). One possible such "technologically progressive monopoly" in the United States may have been Bell Laboratories, owned by AT&T prior to its breakup in 1982. But it can be argued that AT&T was innovative because it perceived the threat of potential competition that eventually arrived.

Intermediate market forms, notably **monopolistic competition** and **oligopoly,** lie between monopoly (one firm) and perfect competition (many firms, none with any control over price). Monopolistic competition involves many firms, each having some price-setting power as a result of product differentiation, such as restaurants. Some customers will remain with the firm when it raises its price because of the perceived uniqueness of its product. In the long run, such firms produce at a lower level of output than the one at which their average costs would be minimized: the **excess capacity theorem.** However, there is little that any government can do about this problem, and none have tried.

Oligopoly, with a small number of firms in the industry, is more complicated. There are many different models of oligopoly behavior because the optimal behavior of an oligopolist will depend on how its fellow oligopolists react to any action it takes. Different reactions imply different outcomes. Many manufacturing sectors subject to substantial economies of scale, such as automobiles and aircraft, fall into this category.

Oligopolistic industries range from very monopolistic to very competitive. At the monopolistic extreme is **perfect collusion,** the joint profit-maximizing cartel. These industries tend to be unstable because a member of the cartel can make extraordinary profits by cheating through price cutting. The oil cartel, OPEC (Organization of Petroleum Exporting Countries), raised oil prices in 1973 and 1979, but in 1986 lost control of world prices as Saudi Arabia increased production to punish Iran and Iraq for cheating on their production quotas (although OPEC has regained some of its market power in more recent years).

Probably the longest surviving cartel in the world is the diamond cartel, based in South Africa. In the United States most cartels are illegal.

At the other extreme, some oligopolistic industries behave like perfectly competitive ones, charging prices almost equal to their marginal costs. These are known as **contestable markets** and are most likely to occur where there are few barriers to entry and exit and the threat of potential, if not actual, competition is ever present. The U.S. airline industry might have been an example prior to facing airport capacity constraints, and Microsoft essentially argued that it behaves like a firm in such an industry in its defense in its antitrust case. This suggests a laissez-faire attitude to oligopolies because they may be reasonably efficient, constantly innovating and investing to keep ahead of potential competitors.

Among market economies, historically the United States has had the most vigorous antitrust policy. The beginning of its policy, and still its most frequently used instrument, was the **Sherman Act of 1890,** which forbids "combinations in restraint of trade" and "efforts to monopolize interstate trade." The Sherman Act was supplemented by the Clayton Act of 1914, which forbids monopolistic stock mergers, interlocking directorates, tying contracts, and price discrimination; the Federal Trade Commission (FTC) Act of 1914, which forbids false advertising; and several later laws as well. The FTC and the Antitrust Division of the Department of Justice have been the main antitrust enforcement bodies of the U.S. government.

Since 1890, U.S. antitrust policies have oscillated back and forth between vigorous and lax enforcement. From 1901 to 1920 there was quite vigorous enforcement, from 1920 to 1945 enforcement was relaxed, and from 1945 to 1982 enforcement was more robust. Since 1982 enforcement has been more relaxed, largely because increased competition from foreign imports, especially from Japan, has reduced the need for enforcement, although there was some increase in vigor in the 1990s, symbolized by the Microsoft case during the Clinton administration, with more of a relaxation again in the administration of George W. Bush.

Controversy exists regarding the actual time path of industrial concentration in the U.S. market share, although the ownership of assets by the very largest firms appears to have increased some over time. But taking into account foreign competition, competitiveness has probably also increased.

Generally, the United States has a more competitive and less concentrated economy than other countries. This is because of its sheer size—it can support more firms in most industries than can many smaller economies. But this also reflects the stronger tradition of U.S. antitrust enforcement. A limited cross-country comparison is shown in table 2-1.

The major anomaly in table 2-1 is the apparently lower degree of concentration in Japan than in the United States. In Japan, many firms that are officially independent have very close relationships with other firms through the "family of companies" groupings known as

Table 2-1
Industrial Concentration Compared

Country of Top Ten Firms	Average Size Top Ten Firms	Percent Employment Concentration Ratio	Average Three-Firm Concentration Ratio
United States	310,554	13.1	41
Japan	107,106	7.3	n.a.
W. Germany	177,173	20.1	56
United Kingdom	141,156	23.1	60
France	116,049	23.2	66
S. Korea	54,416	14.9	n.a.
Canada	36,990	15.3	71
Switzerland	60,039	49.4	n.a.
Sweden	48,538	49.4	83

Source: The first column shows the average number of employees in 1985 in the 10 largest firms, and the second column is the percent of industrial employment in those firms in 1985; both figures from Frederic M. Scherer and David Ross, *Industrial Market Structure and Economic Performance* (Boston: Houghton Mifflin, 1990), p. 63. The third column shows simple average of market share held by the three largest firms in 12 industries in 1970 and is from Frederic M. Scherer, M. Alan Beckenstein, Erich Kaufer, and R.D. Murphy, *The Economics of Multi-Plant Operations: An International Comparisons Study* (Cambridge, Mass.: Harvard University Press, 1975), pp. 218–219 and 426–428.

keiretsu.[8] South Korea has similar groupings known as **chaebol.**[9] Both Japan and South Korea have higher degrees of concentration relative to the United States than table 2-1 indicates.

A final remark on monopoly is that it is a serious problem in the transformation of many of the former command socialist economies into market capitalist ones. Many industries in the former command socialist economies were state-owned monopolies. Eliminating central planning and controls on prices has allowed these firms to behave monopolistically, with resulting aggravation of inflationary tendencies, unresponsive output, and rising resentment by consumers. Furthermore, many of these firms are so big that shutting them down threatens social upheavals from high unemployment.

Externalities

Another source of inefficiency in a laissez-faire equilibrium can be externalities. These are either costs or benefits that are borne by or accrue to an agent other than the agent generating them. External costs are **negative externalities,** the most controversial one being environmental pollution. External benefits are **positive externalities,** such as technological invention without patent protection for inventors.

8. See chapter 6 of this book.

9. See chapter 19 of this book.

To illustrate the effects of externalities, consider the case of a firm that generates pollution damaging another industry but does not reduce that damage if it does not have to pay for it. The private marginal cost to the firm does not equal the social marginal cost and too much pollution is produced, resulting in inefficiency. If an inventor has no patent protection, then other firms can steal her invention and she may make no money even if her invention generates great social benefits. Private marginal benefit to the inventor does not equal marginal social benefit of the invention and too little inventing will occur, resulting in inefficiency. The case of a negative externality is depicted in figure 2-4.

For the efficiency theorem, the problem raised by the existence of externalities, either positive or negative, is an **incompleteness of markets.** For unaccounted-for pollution there is no market for environmental quality, even though environmental quality is something that people desire and that provides utility for them. It has long been argued that the solution to the problem of pollution, or of externalities in general, is to "internalize" them, that is, to make sure that those generating the externalities either bear the costs or receive the benefits they generate.

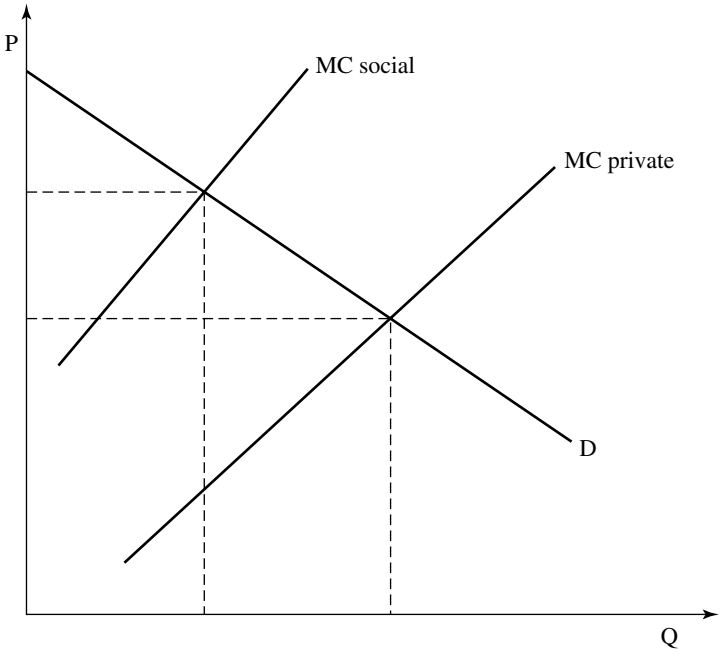


Figure 2-4
Negative externality.

Four broad approaches to resolving the problem of externalities have been proposed within market economies. The earliest approach was provided by A.C. Pigou,[10] who suggested taxation of negative externalities and subsidies of positive externalities. For pollution, the tax equals the difference between the marginal private cost and the marginal social cost, as shown in figure 2-4. There will be a higher price for the commodity whose production generates the pollution and a reduction of its production. The people who consume the polluting commodity ultimately bear the cost.

This taxation strategy has been little used in the United States. Countries using it include France and Germany, with the thrust of their policies being not to raise the cost so high as to discourage polluters from polluting but to raise money for subsidy programs that pay polluters to clean up.

A second policy, widely used in the United States, is command and control quantitative or technological restrictions. In terms of figure 2-4 an efficient strategy moves the equilibrium from the private one to the social one. The optimal size of such a move will vary from industry to industry and from region to region. The social costs of pollution are higher in New Jersey than in North Dakota because of the greater number of people affected by pollution in New Jersey. Nevertheless, until recently, the tendency has been to apply the same emissions or technology standard for a given industry everywhere. This is inefficient, although it might be justified on the grounds of minimizing administrative costs or on some "fairness" notion of everyone doing the same thing.

A third approach takes a more laissez-faire attitude and emphasizes the clear definition and enforcement of property rights. This approach derives from the **Coase Theorem.**[11] This states that if property rights are well defined and transactions costs (or negotiation costs) are negligible, then externalities will be automatically internalized by a market capitalist economy. Ronald Coase presents the example of a railroad whose trains generate sparks that start fires on property adjacent to the railroad's tracks. By mutual negotiation a solution is worked out, such as the railroad compensating the property owners or buying their property.

When property rights are poorly defined and a natural resource is an open-access, common property resource, such as fisheries in international waters, there is a tendency for the resource to be overexploited. No one accounts for the effects of his behavior on others using the resource, so a gap between private and social costs emerges. For fisheries such overexploitation leads to the collapse of fish populations, as has happened to many species. But the essential issue is that of open access rather than common property. Thus the Soviet

10. A. C. Pigou, *The Economics of Welfare* (London: Macmillan, 1922).

11. Ronald Coase, "The Problem of Social Cost," *Journal of Law and Economics* 3 (1961): 1–44. This article is often cited as a foundation of the New Institutional School, which emphasizes the role of transactions costs in how economies are organized.

Union managed the Caspian Sea caviar fisheries well by controlling access, but after its dissolution there was drastic overfishing and a collapse of caviar production because newly independent republics bordering on the Caspian Sea all wanted access.

The Coase Theorem implies serious limits: Property rights may be impossible to define, and negotiation costs may be very high. These conditions are likely to coincide when the externality involves an inherently collective good "owned" by large numbers of people, such as the global air quality and global climate in discussions about global warming. As the inability of world leaders to agree on how to implement the Kyoto Protocol on global warming demonstrates, negotiations to resolve disagreements are far from costless.

Awareness of the Coase Theorem has stimulated the search for market mechanisms to use where possible to resolve pollution problems. An innovation now used widely in the United States is **tradeable emissions permits.** These involve some level of government setting an overall quantitative limit for the emission of a particular pollutant for a particular area and then issuing permits for emissions adding up to that total limit that firms may buy and sell from each other, thereby creating an artificial market for pollution cleanup. The U.S. Clean Air Act of 1990 successfully relied on this approach to deal with sulfur dioxide (SO_2) pollution.

Marketable emissions permits directly attack the problem of *incomplete markets* underlying the inefficiency of unresolved externalities. As long as there are enough parties to make such a market reasonably competitive, it will achieve the least costly solution to cleaning up the given amount of pollution. Those firms that can clean up cheaply do so and sell their "permits to pollute" to those that cannot. One form of this program involves offsets as described in box 2-1.

Active efforts to deal with environmental problems originated in the highest-income countries first and generally in the market capitalist countries before the command socialist countries, despite the theoretical ability of the latter to plan to avoid pollution. This may be more a result of the greater democracy of the advanced market capitalist economies than of their respective economic systems.

The level of development of an economy and its amounts and kinds of pollution seem to be strongly related. Poorer countries seem to have worse water pollution and more particulate matter in the air. Richer countries tend to emit more greenhouse gases, especially carbon dioxide (CO_2) from fossil fuel burning. Middle-income countries tend to emit the most SO_2 responsible for acid rain, largely as a result of burning high-sulfur coal.

Data on different air pollutants in different countries are shown in table 2-2. CO_2 per capita shows a strong correlation with income levels, reaching a peak in some oil-rich Persian Gulf states, although with the United States not too far behind. Nitrogen oxide (NOX) comes largely from automobiles and also shows an income relation, with more pollution from market capitalist economies with more cars; the United States is at the top of the list of per

Box 2-1
Trading in Offsets in the United States.

The U.S. Environmental Protection Agency has set upper limits on the emissions of particular pollutants within certain regions. Thus, if a region is at its limit and a new business wishes to open that would put the region over the limit, it must negotiate a deal with an existing business to reduce emissions by the requisite amount. Such a deal is known as an *offset* and has been used increasingly to maintain air quality without hampering economic growth. Examples include the following:

1. The Narragansett Electric Company has two generating stations in Providence, Rhode Island. Under the bubble policy they were allowed to use high-sulfur oil (2.2% sulfur) at one plant when the second plant was burning natural gas or was not operating, instead of being required to burn 1% sulfur at both plants. This action resulted in a savings of $3 million annually, reduced the use of imported oil by 600,000 barrels per year, and reduced sulfur emissions by 30%.

2. Manufacturers of cans were allowed to comply with existing regulations for each individual can-coating line by averaging emissions of volatile organic compounds on a daily basis, so long as the source did not exceed the total allowable plantwide emission per day. This was expected to save the industry $107 million in capital expenditures, $28 million per year in operating costs, and 4 trillion BTUs of natural gas per year, chiefly because add-on pollution control equipment that would have been energy-consuming is no longer necessary.

3. A cement company in Texas entered into an agreement with another local company providing for that company to install dust collectors. The cement company paid for the equipment while the other company agreed to accept the maintenance costs, which were negligible.

4. A company wanting to build an oil terminal to handle 40,000 barrels a day in Contra Costa County, California, was granted a permit when it acquired, for $250,000, an offset created when a local chemical company shut down.

Source: Tom Tietenberg, *Environment and Natural Resource Economics,* 6th ed. (Boston: Addison-Wesley, 2003), p. 378.

capita output. SO_2 comes from burning dirty coal and oil; per capita emissions are highest in heavily industrialized former command socialist countries, although the United States has the highest levels among the market capitalist countries. Although the United States shows some improvement for SO_2, it shows little for either NOX or CO_2. The former command socialist economies experienced substantial reductions in SO_2 emissions between 1989 and 1996 that go beyond what could be explained by reduced GDP levels.

Collective Consumption Goods

Another source of inefficiency for laissez-faire equilibrium can be **collective consumption goods,** also known as *public goods.* The source of the inefficiency problem with such goods in laissez-faire economies is again incomplete markets. Because of the nature of such goods, it is difficult for private markets to organize themselves to provide these goods in optimal quantities. Thus it often falls to the public sector to do so, leading to inefficiencies associated with that sector. National defense is an archetypal collective consumption good, and the cost overruns and excessive spending of the U.S. defense establishment are infamous.

Table 2-2
Pollution Emissions in Various Countries

	CO_2 (per Capita)		NOX (per Capita)		SO_2 (per Capita)	
Country	1989	1996	1989	1996	1989	1996
Niger	0.15	0.12	n.a.	n.a.	n.a.	n.a.
Egypt	1.54	1.54	n.a.	n.a.	n.a.	n.a.
Costa Rica	0.88	1.28	n.a.	n.a.	n.a.	n.a.
Mexico	3.70	3.75	n.a.	n.a.	n.a.	n.a.
China	2.16	2.73	n.a.	n.a.	n.a.	n.a.
India	0.77	1.05	n.a.	n.a.	n.a.	n.a.
Laos	0.07	0.07	n.a.	n.a.	n.a.	n.a.
S. Korea	5.20	9.00	n.a.	n.a.	n.a.	n.a.
Japan	8.46	9.28	n.a.	n.a.	n.a.	n.a.
Iran	3.11	4.20	n.a.	n.a.	n.a.	n.a.
Saudi Arabia	12.79	14.22	n.a.	n.a.	n.a.	n.a.
Qatar	37.59	n.a.	n.a.	n.a.	n.a.	n.a.
Albania	3.06	0.62	2.8	n.a.	15.6	n.a.
Bulgaria	11.87	0.65	16.7	11.5	114.6	80.6
Czech Republic*	14.47	12.28	60.7	35.3	178.9	90.2
Hungary	6.05	5.83	24.5	20.2	115.2	76.8
Poland	11.54	9.23	39.1	35.2	103.3	76.2
Yugoslavia†	5.61	3.41	8.0	6.9	69.6	59.5
Russia‡	13.26	10.68	14.6	10.0	32.4	19.5
E. Germany§	10.48	10.51	42.6	n.a.	313.3	n.a.
W. Germany	10.48	10.51	48.4	n.a.	24.2	n.a.
Austria	6.82	7.36	27.7	23.3	16.3	6.3
France	6.38	6.21	30.1	31.2	27.1	21.5
Sweden	7.00	6.13	35.4	27.7	25.9	18.1
United Kingdom	9.89	9.53	43.9	33.2	62.1	33.6
United States	19.68	19.67	79.6	78.3	83.2	68.7

*Figures for 1989 are for the combined Czechoslovakia.
†Figures for 1989 are for the former Yugoslavia.
‡Figures for 1989 are for the Soviet Union.
§Figures for CO_2 for the two Germanies are for Germany overall for both years. There are no figures for the separate parts for 1996, but the NOX figures in 1996 are 70 percent of those in 1989 and the SO_2 figures in 1996 are only 29 percent of those in 1989 for Germany overall. The especially sharp decline of the latter almost certainly reflects a major decline in what had been very high emission levels in the former East Germany after unification.
Source: Figures for 1989 are from World Resources Institute, *World Resources, 1992–93* (Oxford: Oxford University Press, 1992), pp. 64–65 for NOX and SO_2 (which are in kilograms) and pp. 346–47 for CO_2 (which is in metric tons). Figures for 1996 come from World Resources Institute, *World Resources 2000–2001* (Washington, D.C.: World Resources Institute, 2000), p. 284 for NOX and SO_2 and pp. 282–283 for CO_2, based on certain calculated adjustments.

The characteristics of a pure public good are *nonexcludability of consumption* and *nondepletability of consumption;* the former is more crucial. These two characteristics together imply that the good is collective in nature; everyone consumes it simultaneously, and no individual's consumption reduces any other individual's consumption. National defense is a classic example and is almost universally provided publicly, even in strongly market capitalist economies. If one individual is defended from foreign invasion then all individuals are so defended, irrespective of whether or not they paid for it.

The essential problem for private market provision of true collective consumption goods is the **free rider problem.** The collective nature of the good breaks the link between paying for it and consuming it. If it exists, everyone consumes it whether or not he or she paid for it. Thus, if the private sector were to attempt to provide the good, it would have a great deal of trouble selling the good because of free riders who would consume it without paying for it. Even though people want the good, they will not pay enough for it to be provided to the Pareto optimal level. This may lead to government taxation to provide the good.

One criticism of this view comes from a radical version of the philosophical perspective known as **methodological individualism,**[12] associated with the mostly pro-laissez-faire Austrian School. An extreme version of this view is in the *objectivist* philosophy of novelist Ayn Rand, who argued that there is no such thing as a human collectivity; ultimate reality is individual people. Apparent collectivities are illusions created to subject individuals to arbitrary tyrannies. However, at least Friedrich Hayek of the Austrian School recognized the existence of some minimal public goods, notably the constitutional maintenance of basic law and order for the protection of property rights and the functioning of free markets.[13]

Between the extremes of pure private goods such as food and pure public goods such as basic law and order, there is a wide spectrum of intermediate goods that have both private and collective aspects. One example is education, which is provided by a mix of public and private sources in the United States, with the public more prominent at lower educational levels and the private gaining in significance at higher levels. Widespread literacy and elementary education of the populace have a significant collective component element for citizens behaving within society. At higher levels of education individuals are more able to appropriate for themselves the benefits of their education, although there are still broader spillovers.

This broad spectrum allows room for variation across societies, even among largely market capitalist economies, regarding the public versus private provision of such intermediate

12. Methodological individualism implies analysis of society from the viewpoints of individuals. This is not inconsistent with the existence of public goods, as their possible existence can be demonstrated from its perspective.

13. There are divisions within the Austrian School, with followers of Friedrich Hayek more willing to accept government action, in contrast with the more firmly pro-laissez-faire followers of Hayek's mentor, Ludwig von Mises, such as Murray Rothbard.

goods. The recent movement in many countries to privatize previously public activities highlights this debate, with no clear criteria regarding optimal outcomes.

Compared to most other market economies, the United States has more of these intermediate activities provided privately. Nevertheless the U.S. economy has a substantial public sector, the most rapidly growing part of which has been at the state and local levels. At the federal level there has always been substantial government ownership of land, especially in the western states. Overall public ownership of land and also of structures is on the order of 15% of the respective totals. These figures represent moderate declines from 1939, which ended a period of increasing government ownership. Areas with significant public participation in the United States include law and order, national defense, the National Forest Service and National Park Service, major dams, the space program, the Tennessee Valley Authority, and numerous local public services such as education, fire protection, local transportation, airports, harbors, highways, garbage collection, water, sewage disposal, libraries, and even some locally owned utilities. Most of these are subject to debate as to whether or not they should be privatized.[14]

Finally, the **public choice school**[15] observes that decisions regarding private versus public ownership are made by legislative bodies at any level of government. These bodies are subject to complexities of majority rule, logrolling, lobbying by special interest groups, and sheer inertia, suggesting that they are ill suited to efficient decision making regarding the proper balance between the private and public sectors. Even though there is a case for public provision of collective consumption goods, the public choice school sees the public sector as so inefficient and corrupt that generally privatization is the preferred market solution.

Imperfect Information

Of all the assumptions needed for the efficiency of an equilibrium outcome, that of perfect information is the most unrealistic. There is no perfect information anywhere about anything, but special problems arise when one party in a transaction knows more than another; this is known as **asymmetric information.**

George Akerlof analyzed the used-car market, in which there is asymmetry of information between the owner of the used car, who knows its flaws, and the potential customer, who does not.[16] But the potential customer understands this disparity and therefore is suspicious of all used cars, suspecting them to be "lemons" their owners wish to dispose of. The inefficiency exists for anyone who seeks to sell a used car that is *not* a lemon at a decent

14. In some cases, partial privatization is what has happened, as with the Federal National Mortgage Association (Fannie Mae), which is involved in the secondary mortgage market.

15. Also known as the Virginia School, its founders were James Buchanan and Gordon Tullock.

16. George Akerlof, "The Market for Lemons," *Quarterly Journal of Economics* 84 (1970): 488–500.

price. Recognition of this by potential sellers simply reinforces the situation; those with good cars who do not *have* to sell won't.

Such asymmetries are rife in market economies. In contractual relationships this is known as the **principal-agent problem,** in which someone is hired who does not do what is best for the employer because of his ability to mislead the ignorant employer. Such information asymmetries can lead to suboptimizing behavior because of **moral hazard,** especially in the insurance industry. Those most needing insurance will seek it out and will conceal their need from the insurers, thereby raising rates for those who need insurance less. Those who are insured then may behave more carelessly than they would if they were uninsured. The U.S. Savings and Loan crisis was an example of a situation in which financial institutions engaged in reckless lending practices, as depositors were not scrutinizing the institutions' behavior because their deposits were insured by the government.

There is no easy way out of the dilemmas posed by imperfect information and asymmetries of information. However, a possible remedy available to government is simply to increase the amount of relevant information generally available. This constitutes an economic efficiency justification for government data-gathering agencies such as the Bureau of Labor Statistics and the Census Bureau. A trial balloon to privatize information gathering and dissemination by the U.S. National Weather Service that was floated during the Reagan administration created a storm and was rejected, showing the strength of support for such activities to be carried out by the public sector.

Another perspective is that the essence of markets is information transmission about relative scarcities through price signals. Hayek argues that a central planner can never possess adequate information for carrying out optimal or even remotely intelligent planning. Free capitalist markets may suffer from imperfect information, but according to this view, they beat command socialist planning in this regard. Decentralized and profit-motivated market capitalists respond to price signals to move the economy along optimally, even though the individual actors possess only limited knowledge.[17] But ultimately, the problems of imperfect and asymmetric information remain unresolved for all economic systems.

THE ROLE OF LABOR UNIONS

Another issue arising in market capitalist economies is the organization of labor into unions. If labor markets are perfectly competitive and lacking in discrimination on characteristics irrelevant to productivity, unions may not contribute to economic efficiency. They

17. That prices may not always accurately transmit information is shown by the example of speculative bubbles, in which prices rise because agents expect them to rise. Some argue that the instability of prices is so endemic as to render market capitalism more generally dysfunctional (Michael Perelman, *The Natural Instability of Markets: Expectations, Increasing Returns and the Collapse of Markets* [New York: St. Martin's Press, 1999]).

redistribute income to their members and deal with safety, job security, benefits, social functions, and political lobbying for broader social outcomes. If they do not offset the monopsonistic power of big firms doing the hiring, their exercising monopolistic power in supplying labor may result in inefficiency.[18]

Although craft unions resemble the medieval European guilds, modern labor unions arose from the revolutionary socialist movements in the nineteenth century, even though independent labor unions were snuffed out by actually ruling command socialist regimes once in power. The Grand Consolidated Trade Union of Britain in the 1830s and the Knights of Labor in the United States in the 1870s were such unions, devoted more to the idea of a general working-class revolution than to negotiating bread-and-butter issues. Given these roots, labor unions have been weakest in the more strongly market capitalist countries such as the United States.

As the harshness of nineteenth-century capitalism moderated in the twentieth century, labor unions became more legally accepted and moderated their approaches to negotiation and political activism. In the United States, the American Federation of Labor (AFL) was founded in 1881 by Samuel Gompers and eschewed radicalism for **business unionism.** This group consists of unions of workers in craft unions, organized like guilds by skill categories (carpenters, plumbers, etc.). Although in Europe acceptance of unions came earlier, the environment in the United States remained fundamentally hostile until the New Deal of Franklin D. Roosevelt passed the Wagner Act of 1935, which established the modern U.S. system of collective bargaining.

The Wagner Act triggered a period of intense organizing and a conflict between the AFL and the newly formed Congress of Industrial Organizations (CIO). In the CIO, unions were organized on industrial lines such as the United Auto Workers and the United Steel Workers, which were more radical and had less-skilled workers. In 1947 a reaction set in with the Taft-Hartley Act, which put restrictions on union activities. By 1955 the percentage of U.S. workers in unions had peaked and the two federations united as the AFL-CIO. Their base in heavy industries means that their membership and influence have declined in the United States as its economy has evolved toward a postindustrial service pattern.

In contrast, Western European countries have larger proportions of their labor forces as union members; the unions are more centrally organized, are more accepted by the political establishment, and are more influential in pushing for greater government intervention in their economies. They serve as the main political-economic base for Western European social democracy.

In some countries labor-management relations have been cooperative, as in Sweden and Germany, whereas in others they have been contentious, as in France and Great Britain. In

18. The original applications of the Sherman Antitrust Act in the United States in the 1890s were against labor unions, although they were very weak at the time. The Clayton Act of 1914 exempted unions from antitrust law.

Table 2-3
Percentages of the Labor Force Belonging to Unions

Country	1955	1975	1995
United States	24.7	21.7	14.2
Japan	n.a.	34	24
Austria	63	59	41.2
Belgium	53	66	n.a.
Denmark	59	67	80.1
Finland	31	75	79.3
France	25	25	9.1
West Germany*	34	33	28.9
Italy	n.a.	47	44.1
Netherlands	38	40	25.6
Norway	54	61	n.a.
Sweden	68	82	91.3
United Kingdom	44	51	32.9

*Unified Germany after 1991.
Source: Data for the United States are from *Economic Report of the President* (Washington, D.C.: USGPO, 1990). Japanese figures are from Richard B. Freeman and M.E. Redick, "Crumbling Pillar? Declining Union Density in Japan," *Japanese and International Economies* 3 (1989): 578–601. European figures are from Jan-Erik Lane and Svante Ersson, *Comparative Political Economy* (London: Pinter, 1990), p. 164, and *The European,* November 6, 1997, p. 23.

Japan, union membership is higher than in the United States, but labor-management relations have been more cooperative. When labor-management cooperation involves economywide negotiating under government encouragement for macroeconomic stabilization, this is known as **corporatism.**[19]

Table 2-3 shows percentages of labor forces belonging to unions for 1955, 1975, and 1995 for 13 countries. The U.S. percentage has continued to decline and is now less than 15 percent. The Japanese percentage has declined to 28 percent. There is no relation between the degree of unionization and the degree of radicalism of unions. The two least unionized countries in table 2-3 are the United States and France. In political terms the former has the most moderate unions on the list, whereas in the latter many unions have strong links to the still fairly orthodox Communist Party of France.

Table 2-4 shows days lost to strikes per thousand nonagricultural wage earners and salaried employees for 1985–1989 and 1990–1994 for 15 countries. Generally disputes have declined markedly except in the United Kingdom, Sweden, New Zealand, and especially Denmark. The most disputatious countries were Italy and Canada, and the least contentious one was Switzerland.

19. Originally corporatism was an authoritarian system promoted by fascists, but now it is a more voluntary system found largely in European social democracies.

Table 2-4
Days Lost to Labor Disputes (per 1,000 Workers)

Country	1985–1989	1990–1994
United States	86	43
Australia	227	157
Belgium	52	57
Canada	424	231
Denmark	235	37
France	57	30
West Germany	2	23
Italy	300	240
Japan	5	3
Netherlands	9	16
New Zealand	491	105
Norway	135	72
Sweden	121	57
Switzerland	0	1
United Kingdom	180	37

Source: U.S. News & World Report 120(16), April 22, 1996: 18.

The impact of unionization on an economy depends on many factors. In the United States, unionized workers are paid more than nonunionized workers. This pay differential may have contributed to the weakening of unions as high wages led to unemployment in the auto and steel industries. If unions cooperate with management and support flexibility in labor policies, high unionization may coexist with low unemployment, as in Sweden until recently. Unions may lead to productivity improvements by giving workers a voice. But if unions insist on restrictive laws, such as ones limiting the ability of firms to fire workers (which discourages firms from hiring workers in the first place), then high unionization may lead to high chronic unemployment, as in Belgium.

MACROECONOMIC INSTABILITY OF MARKET CAPITALISM

The General Picture

Although involuntary unemployment of labor due to macroeconomic fluctuations implies that inefficiency exists, the significance of macroeconomic fluctuations goes beyond cyclical unemployment to constitute a central and distinct issue for market capitalism as a system. It is no accident that the popularity of the U.S. Communist Party peaked in the Great Depression. Mass unemployment and unequal distributions of income and wealth, rather than inefficiency, have provided the most pungent propaganda for socialist critics of market capitalism.

Table 2-5
Percentage of Labor Force Unemployed

Country	1921	1931	1951	1961	1971	1979	1993	2000
Australia	5.9	17.9	1.3	2.3	1.8	6.1	10.7	6.6
Austria	n.a.	9.7	3.5	1.8	1.2	1.7	5.7	3.2
Belgium	6.1	6.8	4.4	2.5	1.7	7.1	13.5	10.7
Canada	5.8	11.6	2.4	7.0	6.1	7.4	11.6	6.8
Denmark	3.0	9.0	4.6	1.9	1.1	5.2	12.3	5.6
Finland	1.8	4.6	0.3	1.2	2.2	6.0	n.a.	n.a.
France	2.7	2.2	2.1	1.5	2.6	5.9	11.7	9.2
Germany	1.2	13.9	7.3	0.7	0.7	3.3	7.5	9.2
Italy	n.a.	4.3	7.3	3.4	4.9	7.1	11.3	11.4
Japan	n.a.	n.a.	1.7	1.4	1.2	2.1	2.5	4.6
Netherlands	1.7	4.3	3.2	0.9	2.3	6.6	6.2	2.7
Norway	n.a.	10.2	1.5	1.8	1.5	2.0	n.a.	n.a.
Sweden	1.3	4.8	1.6	1.5	2.5	2.1	9.0	3.9
Switzerland	n.a.	1.2	0.0	0.0	0.0	0.3	4.6	1.9
United Kingdom	11.0	14.8	2.2	2.0	3.8	5.1	10.4	5.3
United States	11.4	15.2	3.2	6.5	5.9	5.8	5.8	4.0

Source: Data for 1921–1979 from Angus Maddison, *Phases of Capitalist Development* (Oxford: Oxford University Press, 1982), pp. 206–208, except for 1993, which are from *The Economist* (August 20–26, 1994), and 2000, which are from *The Economist* (January 20–26, 2001), p. 106.

The major market capitalist economies have been less than perfectly stable over time, as shown in table 2-5 and figure 2-5. The former shows unemployment rates[20] for a variety of countries in various years. There was a general increase in unemployment rates after the early 1970s in many countries, associated with a general stagnation of economic growth, that appears to have been reduced more recently.

Figure 2-5 shows the maximum peak-to-trough decline of GDP over a business cycle for various countries for various periods, a crude measure of instability. There was more such variability before World War II than afterward.

Examination of the components of GDP indicates that the most variable element is capital investment, sometimes characterized as a "flighty bird." The considerable variation in capital investment can be explained by factors ranging from exogenous fluctuations in new technologies that can serve as the basis for investment, to fluctuations in government monetary policies affecting interest rates, to psychological fluctuations due to the "animal spirits" of those making investments. The question is, why do these variables lead to fluctuations in the unemployment rate, since in a perfectly competitive labor market, wage rates should fall when the demand for labor falls, thereby preventing the emergence of any

20. Some variation across countries and over time is attributable to variations in labor force participation rates.

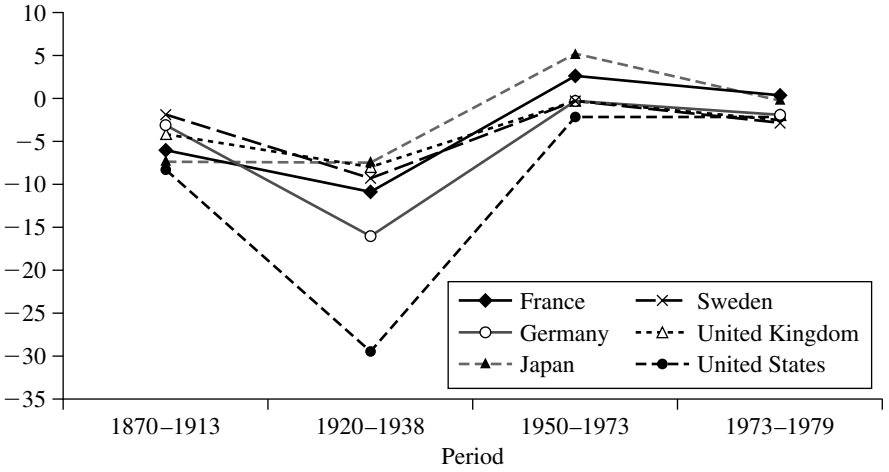


Figure 2-5
Maximum peak-to-trough GDP decline over cycle.
Source: Angus Maddison, *Phases of Capitalist Development* (New York: Oxford University Press, 1982), table 4-1, "Amplitude of Recessions in Aggregate Output, 1870–1979, p. 67.

involuntary unemployment? The **Keynesian School** and the **Classical School** answer this question differently.[21]

Deriving its views from British economist J. Maynard Keynes and his *General Theory of Employment, Interest and Money* (1936), the Keynesian School argues that rigidities of various sorts exist in labor markets and that capital investment can collapse and stay down for extended periods of time, as in the Great Depression. The implication is that government intervention through fiscal or monetary policies is advisable to stimulate the economy and to stabilize and smooth out business cycles. Some Keynesians, such as James K. Galbraith, argue that stimulative macroeconomic policies can also increase the equality of the income distribution.

The Classical School, deriving from nineteenth-century classical political economists such as David Ricardo, argues that market capitalist economies are powerfully self-stabilizing. Conscious government intervention merely generates inflation and aggravates fluctuations. To minimize unemployment, unions should be broken up and a stable fiscal and monetary environment should be maintained within a laissez-faire environment.

21. These two schools have various subcategories including old and new classicals and Keynesians, Post Keynesians, and supply siders. To the left and right of this broad division stand the pro–command socialist Marxists and their ideological opposites, the pro-laissez-faire Austrians.

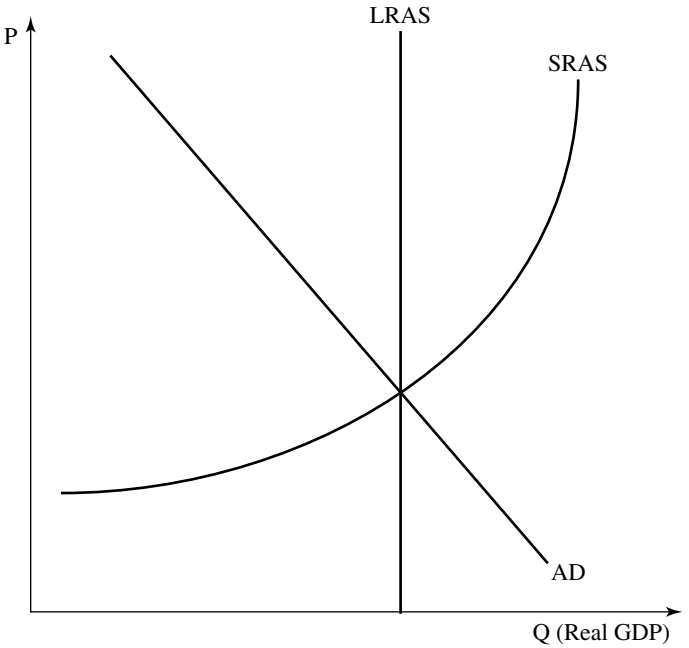


Figure 2-6
Alternative views of macroeconomic equilibrium.

Figure 2-6 depicts alternative views of macroeconomic equilibrium. *P* now represents the overall price level of the economy, and *Q* represents the aggregate level of real output. Aggregate demand (AD) is downward-sloping. However, short-run aggregate supply (SRAS) represents an upward-sloping curve as conceptualized by Keynesians. Its shape implies that a decline in aggregate demand will cause a decline in real output and a decline in employment. Long-run aggregate supply (LRAS) represents a curve as conceptualized by classicals. Most classical economists recognize that output can decline in the short run for a variety of reasons, but they argue that *very quickly* a laissez-faire economy will return to behavior depicted by the LRAS. Declines in aggregate demand will lead to declines in the price level rather than in real output or employment.

Tools of Macroeconomic Policy

Two main macroeconomic policy tools are fiscal and monetary policy. Fiscal policy is a nation's government budget, its package of spending and tax rates. Generally, the higher spending is and the lower tax rates are, the higher will be the stimulus to aggregate demand. Monetary policy is the expansion or contraction of a nation's money supply and the lowering

or raising of interest rates in its economy. Expanding the money supply or lowering interest rates stimulates aggregate demand, while contracting the money supply or raising interest rates dampens aggregate demand.

In parliamentary democracies, where the executive leaders are also the legislative leaders, responsibility for fiscal policy is well defined. In the United States, the separation of the executive and legislative branches of government creates a messier situation. Executive agencies propose spending packages that are filtered through the Office of Management and Budget (OMB), which generally acts as the president's executive agent on spending proposals. The Treasury makes tax proposals. These are combined and submitted to Congress, which chops them up and sends them through a plethora of committees and subcommittees, rarely succeeding in actually passing a complete budget in time for when it is supposed to take effect. Disagreements between Congress and the president then are worked out through negotiation or, in cases of more conflict, through the veto process or even threats of government shutdown, as happened in 1995. What finally emerges and is actually implemented by the executive agencies may not resemble the desired plans of anybody involved. In general, Keynesians favor the use of discretionary fiscal policy for stabilizing macroeconomic fluctuations more strongly than the Classical School, which tends to be skeptical regarding the efficacy of this policy tool.

In all modern market capitalist economies, monetary policy is carried out by a central bank that controls the domestic money supply and at least short-term interest rates. The concentration of decision making in one body generally allows a greater degree of purposiveness and control of monetary policy than exists for fiscal policy. It has been argued that the second most powerful person in the world after the U.S. president is the Chairman of the Board of Governors of the U.S. Federal Reserve System, the U.S. central bank (the *Fed*).

Nations vary considerably in the way their respective central banks relate to the entities in charge of fiscal policy. Some central banks have considerable autonomy and independence, such as the recently created European Central Bank (ECB), which has been modeled on the notably independent Bundesbank of Germany. Others are subordinated to fiscal policymakers, notably the Bank of England, which is administratively under the control of the budget-proposing Chancellor of the Exchequer, although it has been granted effective autonomy since the Labour Party came to power. The U.S. Fed is somewhere in between. It was created by and is subject to rules established by Congress, and its Board of Governors are appointed by the president for 14-year terms. It is widely argued that the more independent central banks have had better records of controlling inflation, and there has been a general global trend toward granting central banks more autonomy.

While the Keynesian School has generally favored discretionary monetary policy responding to recent macroeconomic conditions, with a focus on managing interest rates, the Classical School has tended to favor *monetarism,* the idea that central banks should focus

on controlling the growth of the money supply so that it increases steadily at a rate equal to that of the rate of real GDP growth in order to control inflation.[22] This view peaked in the United States during the 1979–1982 period. More recently, most central banks have focused on managing interest rates, but with the goal of targeting inflation rather than other macroeconomic variables.

A RECAPITULATION OF THE STRONG CASE FOR LAISSEZ-FAIRE

A major theme of the recent world political economy has been a strong shift toward supporting more laissez-faire economic policies. The enormous changes in Eastern Europe have had much to do with this shift, but so have changes in the market capitalist economies. In the former communist world, disillusionment with the old ideology of command socialism has led to great fascination with its extreme opposite, laissez-faire market capitalism. Even the backlash to this system, with former communists winning elections in some countries, has been marked by all of them supporting market-oriented reforms of one sort or another, if only as promises for a more gradual approach to a transition to some form of market capitalism.

There is a tension between asserting the efficiency of competitive equilibria and recognizing the limits of the applicability of that theorem. While recognizing these limits, the stronger advocates of laissez-faire argue against their significance or relevance.

The most straightforward case is made by the **Chicago School,** whose most prominent spokesman is Milton Friedman.[23] The Chicago School's argument draws directly from the efficiency theorem and follows by asserting the irrelevance or unimportance of the various exceptions and limits. Markets are almost always efficient, so government should keep its hands off.

Monopolistic or oligopolistic market structures reflect efficient and competitive behavior by the firms involved unless their behavior is enforced because of some government restriction on entry. Friedman's opposition to government restrictions on entry leads him to oppose the licensing of doctors by government. According to the Chicago School, government-initiated antitrust suits are just a waste of time, distracting firms from their market-appointed task of efficiently providing goods and services to consumers.

22. This policy has been most eloquently and influentially defended by Milton Friedman in *The Quantity Theory of Money* (Chicago: University of Chicago Press, 1957).

23. A sign of Friedman's special prominence is that when the former Soviet leader, Mikhail Gorbachev, visited the Hoover Institution at Stanford University, where Friedman was then located, Gorbachev made a special point of seeking him out for praise and a handshake. His ninetieth birthday was celebrated by a visit to President George W. Bush at the White House. The classic statement of a strong version of Friedman's views is found in his *Capitalism and Freedom* (Chicago: University of Chicago Press, 1962).

The Chicago School argues that most externalities will be resolved by private markets if property rights are properly defined and enforced, as suggested by the Coase Theorem. If they are not, then externalities are probably not very significant and the benefits to be obtained from any possible government intervention will be more than offset by the inefficiencies and waste associated with government activity, it is argued.

The Chicago School of free markets holds that many of the goods provided by the public sector are not really collective consumption goods and could be more efficiently provided privately. Thus privatization of public goods provision should be vigorously pursued.

The Chicago School also argues that information costs are inevitable and cannot be avoided. Again, inefficient governments are unlikely to be very helpful in improving matters in this area.

In macroeconomics the Chicago School supports the Classical School approach. Milton Friedman is the most prominent advocate of monetarism in the United States and has proposed that the money supply should grow at a constant rate per year. Beyond that, fiscal policy should involve low to no deficits and should not be actively used for stabilization efforts.

With respect to the distribution of income, people should be allowed to keep what they earn from the free market. Inequalities are the necessary outcome of providing sufficient incentives for production, investment, and growth. The equity-efficiency trade-off is real, and efficiency should be favored. So, government should not redistribute income.

A more general criticism of government intervention is made by *public choice theory,* which argues that the government agencies designated to carry out the market-correcting activities are self-interested entities that become captured by special interests operating through their legislative connections. This analysis became more focused after the discovery of the concept of **rent-seeking** by Gordon Tullock and Anne Krueger, with *rent* defined as the return to a factor with a fixed supply, like land or a unique individual, that then has monopoly power. Government agencies through regulatory actions can create artificial scarcities, such as a limited number of import licenses in a less developed country. This artificially creates rents that can be captured by special interests, including the bureaucrats in charge of allocating these scarce items by means of bribery. Devoting large amounts of economic resources to the creation and capture of these rents often involves corruption, which has become an increasingly serious problem in many countries (box 2-2).

Finally, we note that the Austrian School generally supports laissez-faire quite strongly, but from a different theoretical perspective. In contrast to the Chicago School's acceptance of the Pareto efficiency theorem of a static Walrasian general equilibrium, the Austrian School rejects equilibrium analysis and emphasizes dynamic market processes in the context of a strictly subjectivist, demand-side theory of value. In their view, entrepreneurs are the most important agents in the economy. They must be allowed to function freely, without

Box 2-2
Does Rent-Seeking Explain National Destinies?

The rent-seeking theory has been broadly applied by Mancur Olson in his *The Rise and Decline of Nations* (1982). He revives the Jeffersonian notion that every society needs a revolutionary shakeup from time to time in order to maintain its dynamism. Without such occasional shakeups, often brought about by defeat or conquest by a foreign power, interest groups accumulate power and influence and are able to expand their rent-seeking activities to the point where the economy loses all ability to change or grow as it becomes encrusted with more and more regulations and restrictions designed to protect the increasingly bloated rents of the outdated special interests. Olson presents the stagnation of never-conquered-since-1066 Great Britain and its ossified class system as a leading negative example and the postwar economic dynamism of defeated Germany and Japan as positive examples. The Olson theory suggests that Russia and its former allies may have bright futures if they can get through their difficult transitions.

government restriction, so that they can lead the market to evolve in conjunction with the evolution of consumer preferences through an ongoing process of innovation. This view draws strongly on the creative destruction ideas of Joseph Schumpeter, although he did not agree with the more laissez-faire views of his fellow Viennese natives of Austria, Ludwig von Mises and Friedrich Hayek. For the Austrian School, static efficiency is relatively unimportant. It is the dynamic success of market capitalism that is its most important attractive economic feature.

SUMMARY AND CONCLUSIONS

Modern market capitalism evolved out of the merchant capitalism of northern Italy and Flanders in the late medieval period, achieving its closest approach to laissez-faire in the Industrial Revolution in Britain and the United States in the nineteenth century. These economies saw increased government intervention in the twentieth century, leaving Hong Kong, Singapore, and New Zealand as the most laissez-faire advanced economies in the world today.

A complete, competitive, full-information general equilibrium, in which supply equals demand in all markets, is efficient in the Pareto sense. Thus no individual can be made better off without making some other individual worse off. This theorem does not address the question of income distribution, and critics of market capitalism emphasize its tendency toward inequalities of income and wealth, which seem to be worsening over time. Supporters of market capitalism, however, argue that such inequalities are necessary to bring about economic growth because they provide incentives for work effort and investment.

The efficiency theorem suggests that a minimal-government, laissez-faire economy might be inefficient because of monopoly power, externalities, insufficient public goods, imperfect information, and possible macroeconomic instability. During the twentieth century

the U.S. economy developed mechanisms for dealing with these problems, many of them involving some sort of government activity or intervention.

Despite this evolution, controversy and debate continue regarding the appropriate scope of government involvement in mixed-market capitalist economies. Strong supporters of laissez-faire critique proposed and actual ways in which governments seek to correct the inefficiencies of unfettered markets. This suggests that as the former socialist countries continue to move toward market capitalism, they will increasingly encounter serious questions regarding the ultimate balance between the public and private sectors within their economies.

Although the United States is much closer to laissez-faire than most market capitalist economies, its government actively intervenes in a variety of ways to deal with the above problems. A recent trend has been toward policies that may involve establishing a market where none existed before, for example, for tradeable pollution permits. Compared with other market capitalist economies, the United States has a very high real income, a high level of competitiveness, greater income inequality, weaker unions, more per capita air pollution, a smaller public sector, and historically somewhat greater volatility of its GDP. Thus it shows the dynamism but also the difficulties of market capitalism as a system.

QUESTIONS FOR DISCUSSION

1. "Income inequality is necessary for Pareto optimality so that people will work as hard as they can." "If income is distributed equally, then there must be Pareto optimality because you can't make one person better off without making someone else worse off." Are these statements true or false? Why or why not?

2. How does the U.S. economy compare with other market capitalist economies with respect to its degree of industrial concentration and the nature of its antitrust policies?

3. Socialist critics of market capitalism argue that businesspeople will not care about the environment because they are concerned only about profits, whereas socialist planning will be able to take care of the environment. Why then have environmental conditions generally been better in the industrialized market capitalist economies than in the industrialized command socialist ones?

4. What are some publicly provided goods in the United States that could be efficiently provided privately instead? Why are they not so provided?

5. Looking at the tables in this chapter, can you find any evidence that the influence of Keynes might have contributed to improved macroeconomic performance among market capitalist economies? How might his influence have failed to improve their performance?

6. How might you critique the strong case for laissez-faire presented at the end of the chapter?

7. How do the defenses of laissez-faire by the Chicago School and the Austrian School differ?

SUGGESTED FURTHER READINGS

Buchanan, James, and Gordon Tullock. *The Calculus of Consent*. Ann Arbor: University of Michigan Press, 1962.

Freeman, Richard B., and James L. Medoff. *What Do Unions Do?* New York: Basic Books, 1984.

Frey, Bruno S. "Decision Making Via the Price System," in Morris Bornstein, ed., *Comparative Economic Systems: Models and Cases,* 7th ed. Burr Ridge, Ill.: Irwin, 1994, pp. 92–114.

Galbraith, James K. *Created Unequal—The Crisis in American Pay*. New York: Free Press, 1988.

Hayek, Friedrich A. "The Price System as a Mechanism for Using Knowledge." *American Economic Review* 35 (1945): 519–530.

Keynes, John Maynard. *The General Theory of Employment, Interest and Money*. London: Macmillan, 1936.

Krueger, Anne O. "Economists' Changing Perceptions of Government," in Morris Bornstein, ed., *Comparative Economic Systems: Models and Cases,* 7th ed. Burr Ridge, Ill.: Irwin, 1994, pp. 78–91.

Maddison, Angus. *Phases of Capitalist Development*. Oxford: Oxford University Press, 1982.

North, Douglass C., and Robert Paul Thomas. *The Rise of the Western World: A New Economic History*. Cambridge: Cambridge University Press, 1973.

Olson, Mancur. *The Rise and Decline of Nations*. New Haven, Conn.: Yale University Press, 1982.

Perelman, Michael. *The Natural Instability of Markets: Expectations, Increasing Returns and the Collapse of Markets*. New York: St. Martin's Press, 1999.

Rothbard, Murray N. *Man, Economy, and State*. Los Angeles: Nash, 1962.

Scherer, Frederic M., and David Ross. *Industrial Market Structure and Performance*. Boston: Houghton Mifflin, 1990.

Schumpeter, Joseph A. *The Theory of Economic Development*. Cambridge, Mass.: Harvard University Press, 1934.

Schumpeter, Joseph A. *Capitalism, Socialism, and Democracy*. London: Allen & Unwin, 1943.

Smith, Adam. *The Wealth of Nations*. London: Strahan and Cadell, 1776.

Tietenberg, Tom, *Environmental and Natural Resource Economics,* 6th ed. Boston: Addison-Wesley, 2003.

von Mises, Ludwig. *Human Action: A Treatise on Economics,* 3rd ed. Chicago: Henry Regnery, 1963.

Williamson, Oliver E. *The Economic Institutions of Capitalism*. New York: Free Press, 1985.